

Project Report: Diabetes Prediction using Machine Learning Models

1. Introduction

This project aims to predict whether a patient has diabetes based on various medical factors such as age, BMI, HbA1c level, and blood glucose level. The problem this project solves is identifying potential diabetic patients early, which can help in preventing complications. The motivation behind this project is to explore how different machine learning models perform in binary classification tasks and compare their performance metrics.

2. Dataset Description

Source

2.1 Link

- The dataset can be accessed [here](#).

2.2 Reference

- Kaggle

2.3 Dataset Description

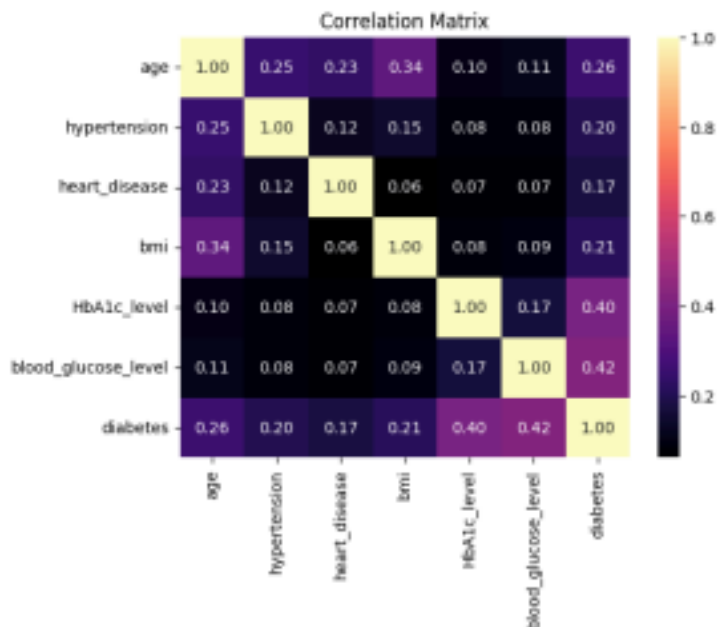
- **Features:** The dataset consists of 9 features and 1 target variable (**diabetes**), which indicates whether the patient has diabetes or not.
- **Classification Problem:** This is a classification problem because the target variable is binary—either the individual is diabetic (1) or non-diabetic (0).

2.4 Features Description

- **Quantitative Features:** **age**, **hypertension**, **heart_disease**, **bmi**, **HbA1c_level**, **blood_glucose_level**, **diabetes**.
- **Categorical Features:** **gender**, **smoking_history**.

2.5 Correlation Analysis

The correlation matrix is used to evaluate the relationship between the features. The heatmap below shows the correlation of each feature with diabetes:



- The highest correlation with diabetes is observed for features like `HbA1c_level` and `blood_glucose_level`.

3. Dataset Pre-processing

3. Data Pre-processing

Identified Issues:

- **Age less than 6 months:**
 - Action: Removed rows with `age < 0.5`.
- **Gender "Other":**
 - Action: Removed rows with `gender = "Other"`.
- **Categorical Features:**
 - Action: Applied label encoding to `gender` and `smoking_history`.

Feature Scaling:

- Applied standard scaling to continuous features (`age`, `bmi`, `HbA1c_level`, `blood_glucose_level`).

Dataset Splitting:

- **Train-Test Split:**
 - 80% for training, 20% for testing.

Exploratory Data Analysis

- **Distribution of Continuous Features:**

Include histograms and density plots for features like **age**, **bmi**, etc.

- **Feature Correlation:**

Correlation heatmap visualizing relationships between features and target variable.

4. Model Training and Testing

4.1 K-Nearest Neighbors (KNN)

- KNN was trained with 5 neighbors, which predicts the class of a data point based on the majority class of its neighbors.
- Accuracy achieved: **96.19%**

4.2 Naive Bayes

- Naive Bayes, which assumes independence between the features, was trained for the classification task.
- Accuracy achieved: **90.10%**

4.3 Logistic Regression

- Logistic Regression was used for binary classification to model the probability of diabetes occurrence.
- Accuracy achieved: **96.12%**

5. Model Comparison and Evaluation

5.1 Accuracy Comparison

The success rates of all models are as follows:

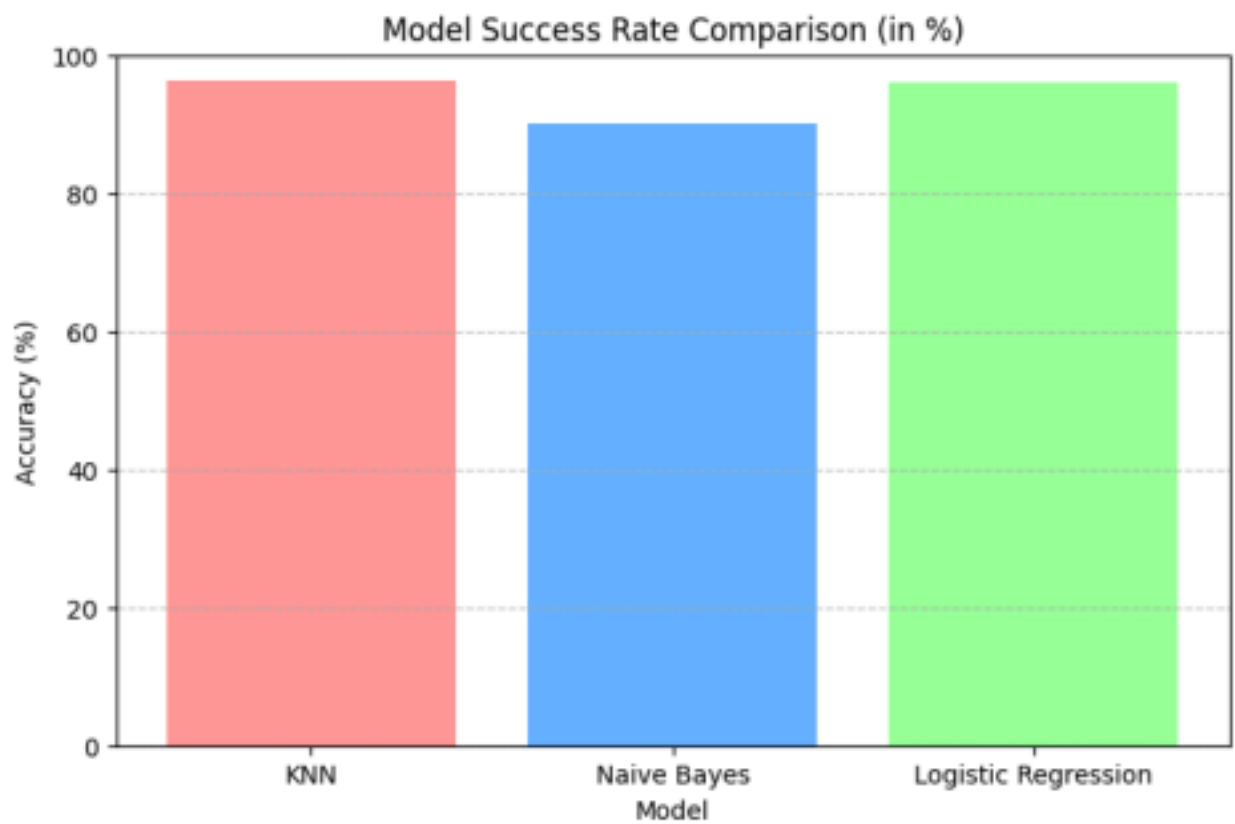
Model Accuracy (%)

K-Nearest Neighbors 96.19

Naive Bayes 90.10%

Logistic Regression 96.12%

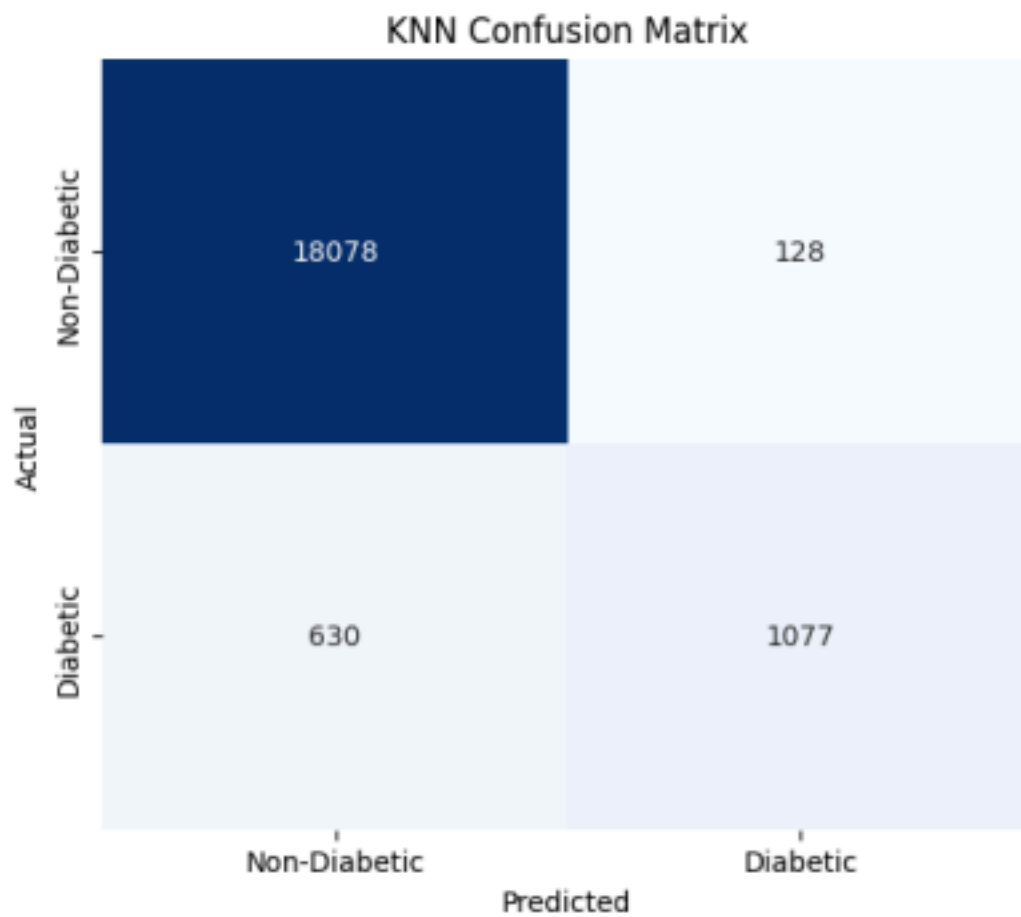
A bar chart comparing the accuracy of the models was plotted:



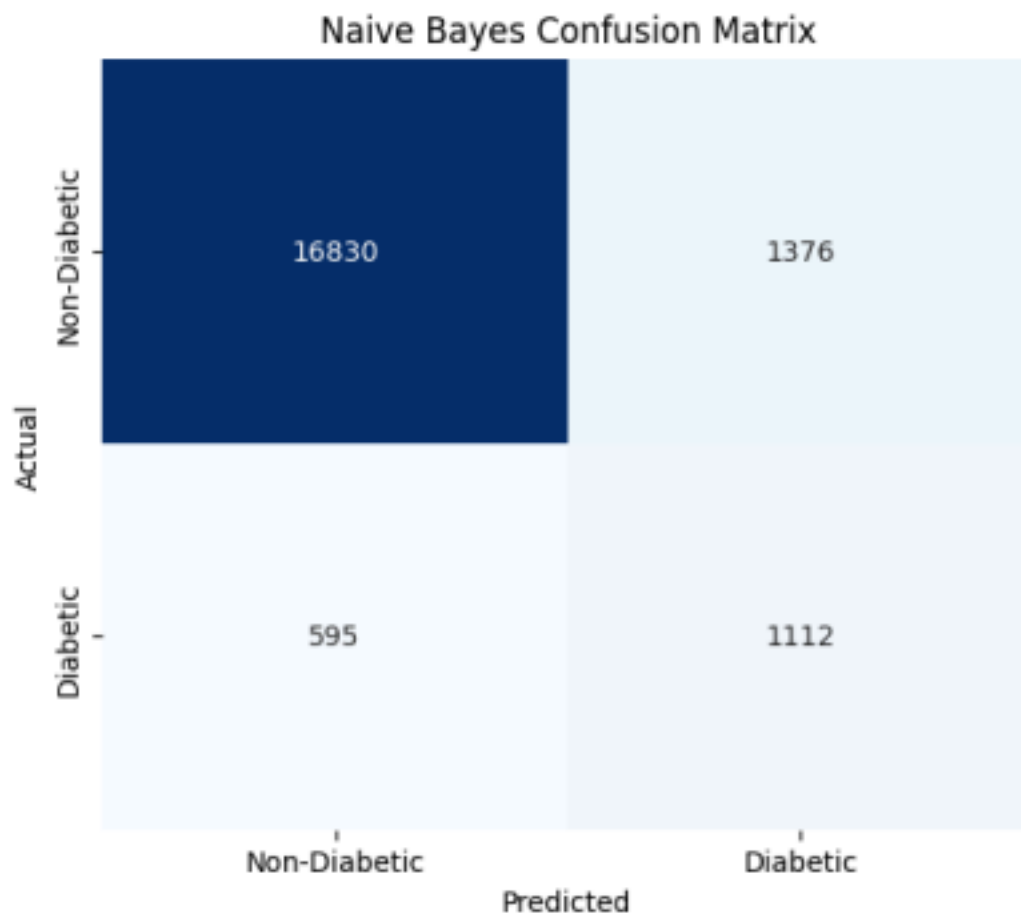
5.2 Precision, Recall, and Confusion Matrix

For each model, we evaluated precision, recall, and plotted a confusion matrix:

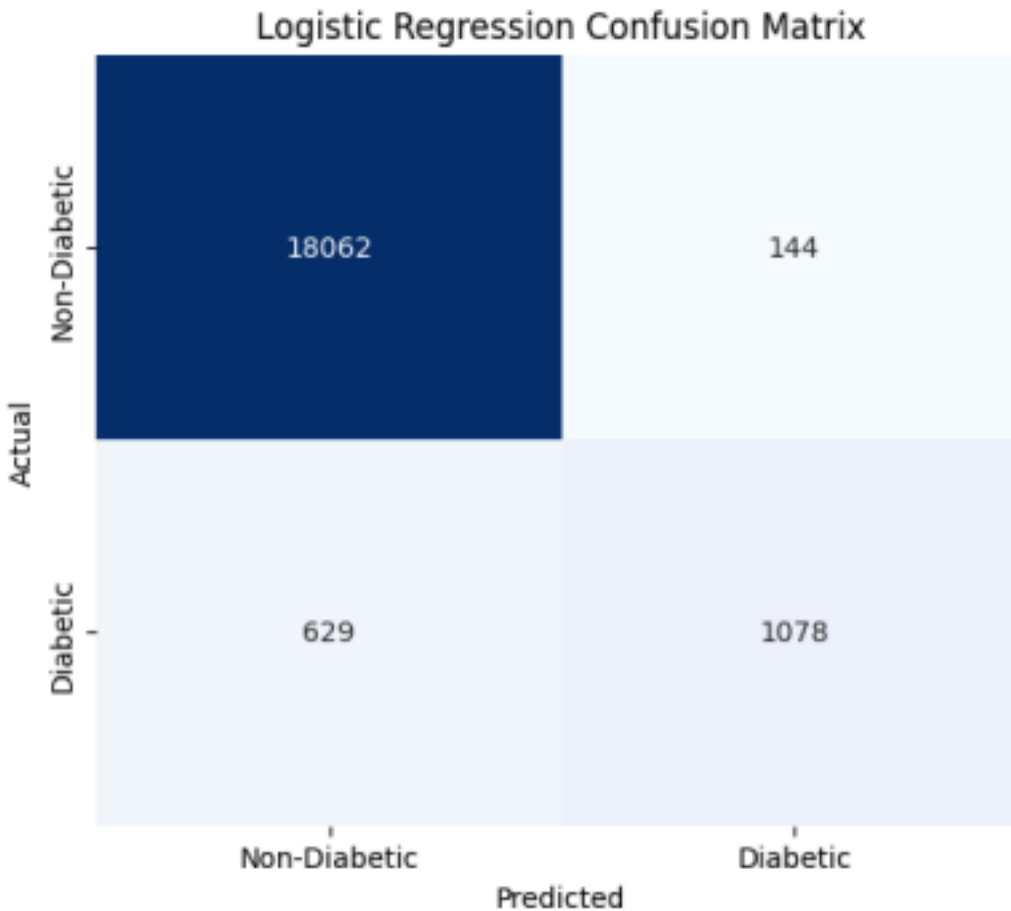
KNN Confusion Matrix:



Naive Bayes Confusion Matrix:



Logistic Regression Confusion Matrix:



- Precision and recall for Logistic Regression were the highest, making it the most reliable model for this dataset.

Key Findings:

- **HbA1c level** and **blood glucose level** were found to be the most important predictors of diabetes.
- KNN demonstrated the highest accuracy, making it the best model for this task.

6. Conclusion

After training and testing several models, KNN emerged as the most accurate model for predicting diabetes in this dataset, with an accuracy of 96.19%. Logistic Regression also performed reasonably well, but Naive Bayes was less effective due to the dependence between certain features.